



BAHIR DAR UNIVERSITY

Department of Software Engineering

Operating system individual assignment

Name- Rediet Kassa

Id-1702510

Section -B

**Submitted to – Mr. wendemu
Academic Year :2026**

Contents

Introduction (Background, Motivation)	3
Motivation:	3
Objectives	3
Requirements	4
i. Hardware Requirements (Host Machine):	4
ii. Software Requirements:	4
Installation Steps	4
File system Support	20
Advantages and Disadvantages	21
Advantages	21
Disadvantages:	21
Conclusion	21
Future Outlook / Recommendation	22
Virtualization	22
What is Virtualization	22
Why Virtualization	22
How does Virtualization Work	23
Key Mechanisms:	23

Installation of Operating System in Virtual Environment

Given Operating System:Endless OS

Virtualization Tool:Oracle VM VirtualBox

Introduction (Background, Motivation)

Endless OS is a Linux-based operating system designed to be simple, beautiful, and accessible, especially for users with limited technical knowledge or internet access. It comes pre-loaded with a large collection of educational and productivity applications, making it an excellent choice for students and first-time Linux users.

Motivation:

The main motivation for downloading Endless OS is its user-friendly interface (similar to a modern tablet/smartphone), offline-first design, and its ability to run smoothly even on modest hardware. Installing it in a virtual environment using Oracle VM VirtualBox allows us to safely experiment with a modern Linux distribution without affecting the host operating system.

Objectives

- To successfully install Endless OS inside a virtual machine using Oracle VM VirtualBox.
- To understand the installation process of a Linux-based OS in a virtualized environment.
- To explore the features, filesystem, advantages, and disadvantages of Endless OS.
- To document the complete installation steps with screenshots.

Requirements

i. Hardware Requirements (Host Machine):

- Processor: Intel/AMD 64-bit CPU with virtualization support (VT-x / AMD-V) enabled in BIOS/UEFI
- RAM: Minimum 8 GB (4 GB allocated to VM recommended)
- Hard Disk: At least 50 GB free space
- Internet connection (recommended for updates)

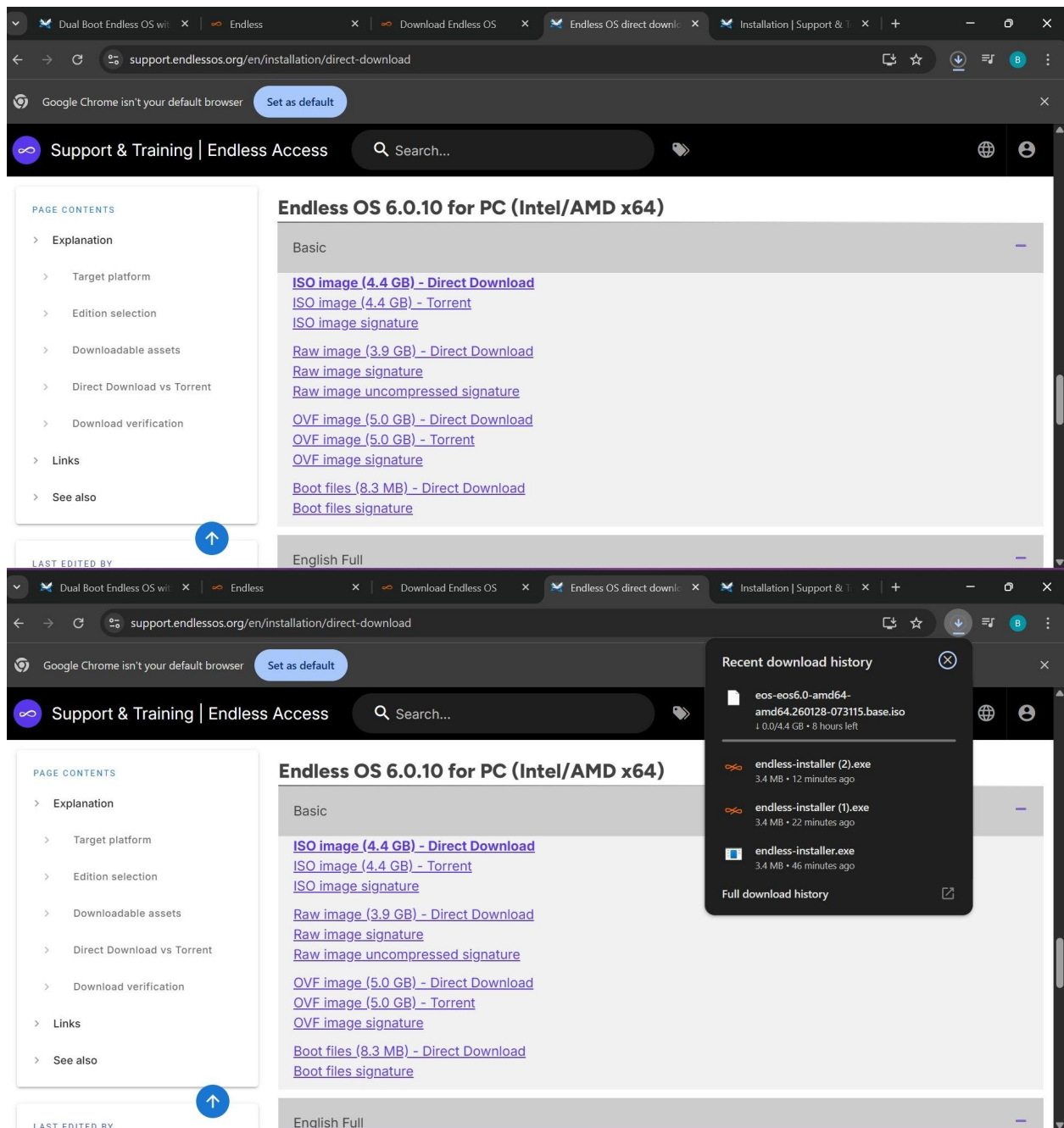
ii. Software Requirements:

- Virtualization Software: Oracle VM VirtualBox (latest version)
- Extension Pack for VirtualBox
- Endless OS ISO image (latest version downloaded from official website: endlessos.org)

Installation Steps

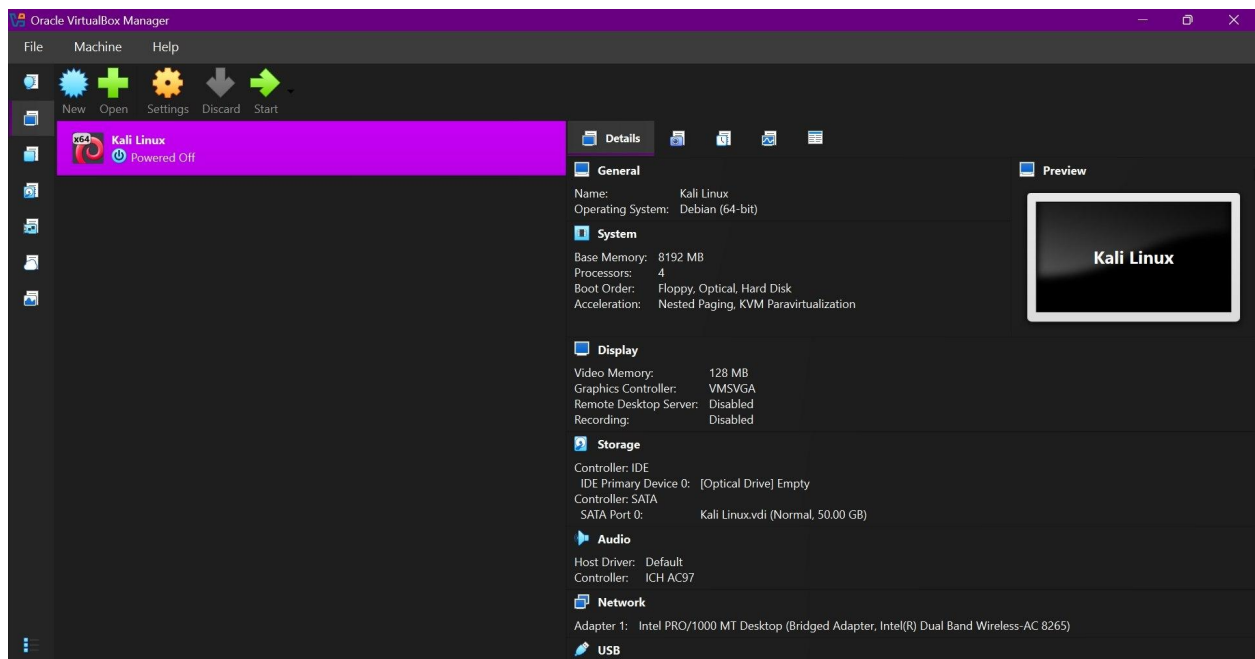
I Download Endless OS ISO

1. Visit the official website (endlessos.org)
2. Download the latest Endless OS ISO file.



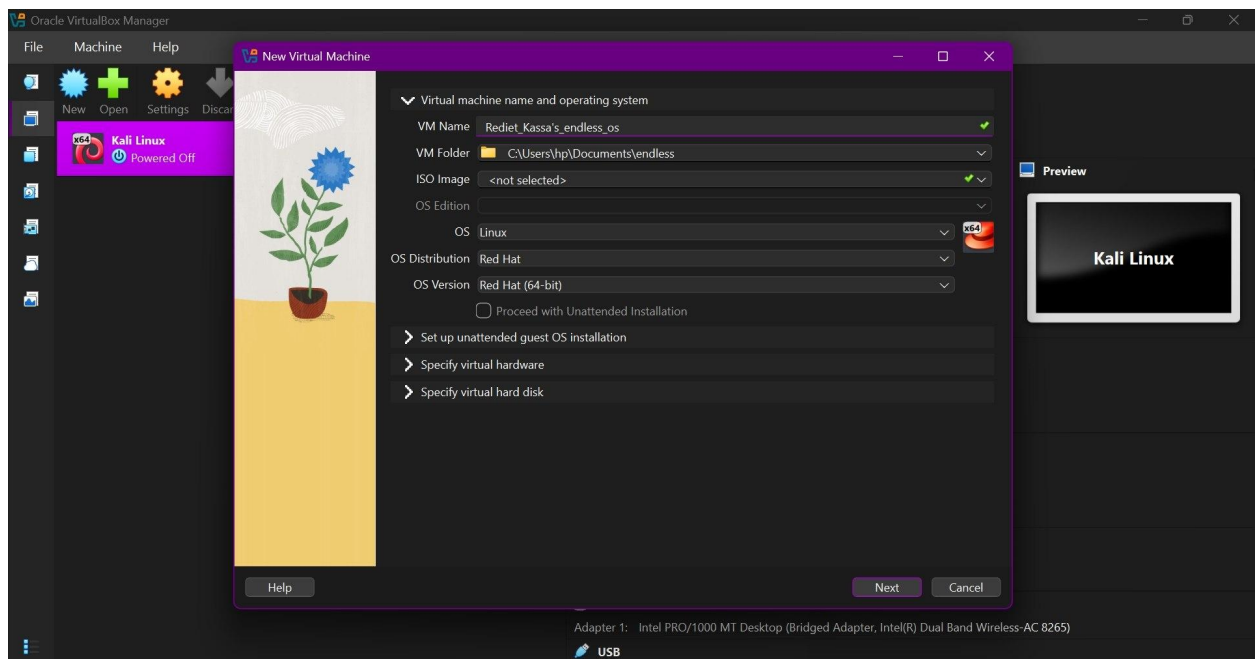
ii. Create New Virtual Machine in Oracle VM VirtualBox

1. Open Oracle VM VirtualBox.
2. Click New to create a new virtual machine.

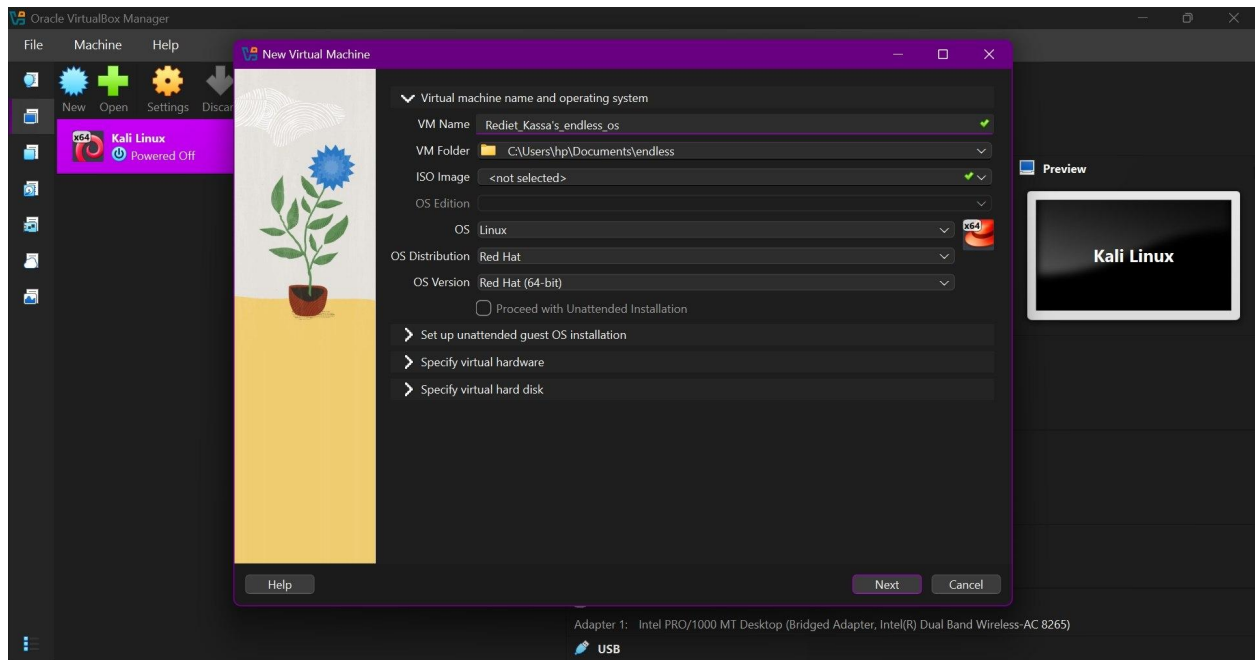


3. Enter the following details:

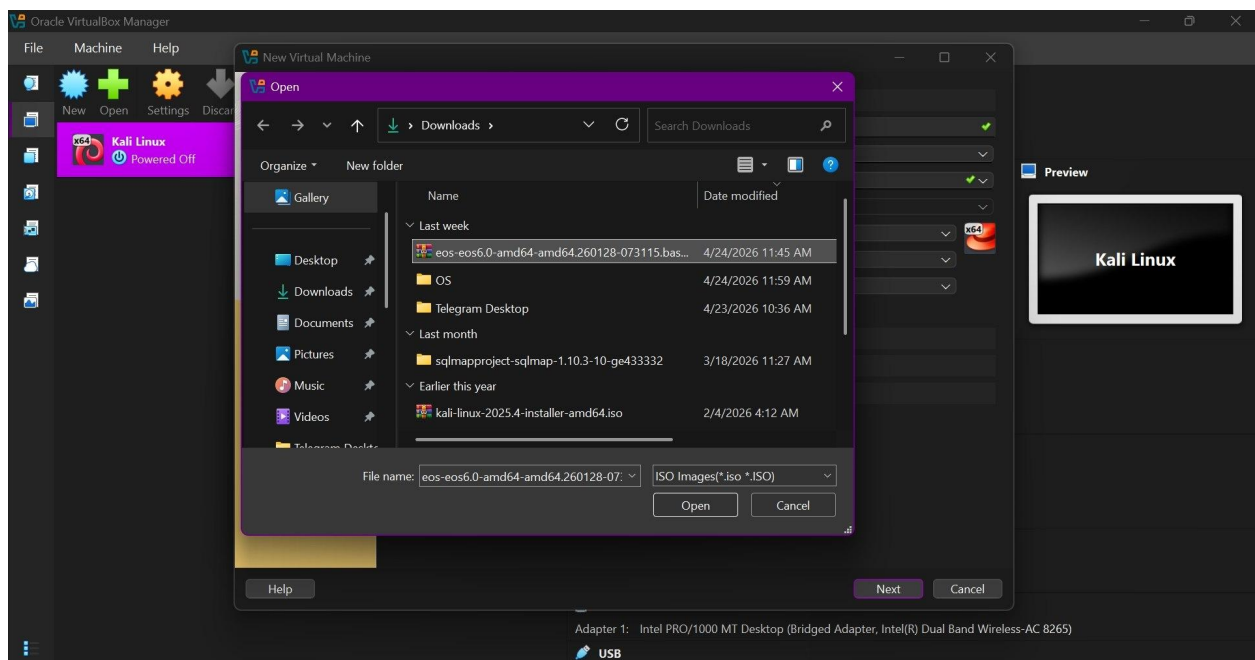
- Name: Endless-OS- Rediet kassa



- Folder: Choose any suitable location



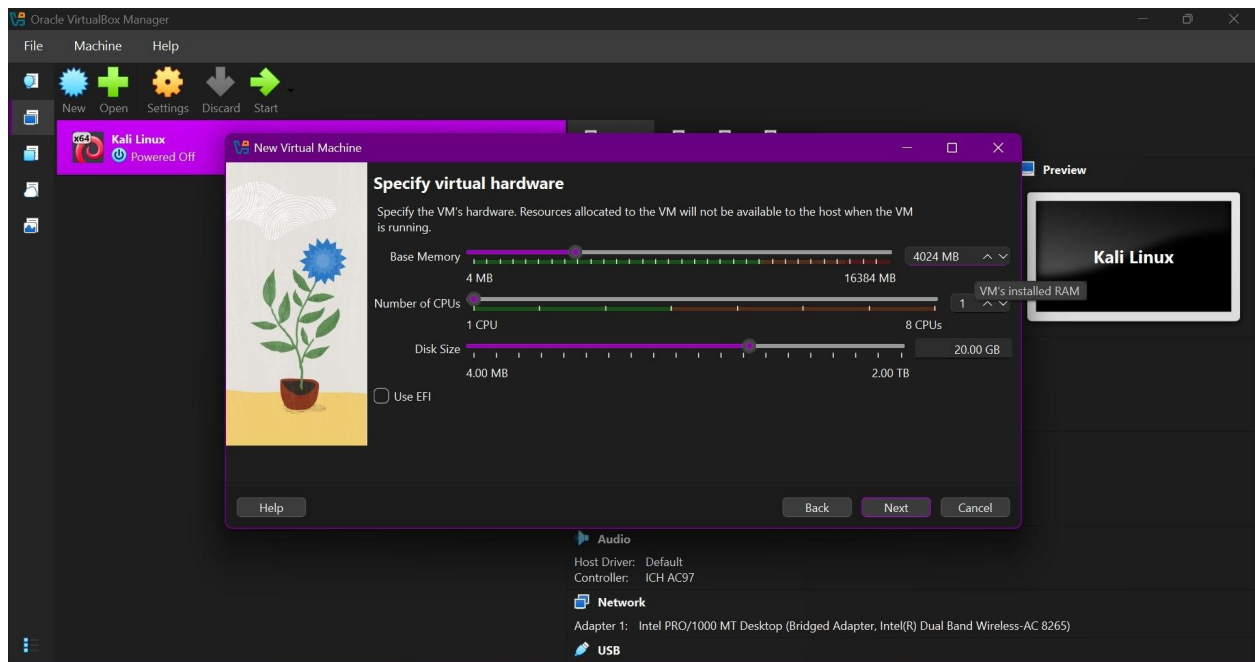
- Version: Other Linux (64-bit)



4. Allocate memory (RAM): 4096 MB (4 GB recommended).

5. Hard Disk: Select Create a virtual hard disk now, Next.

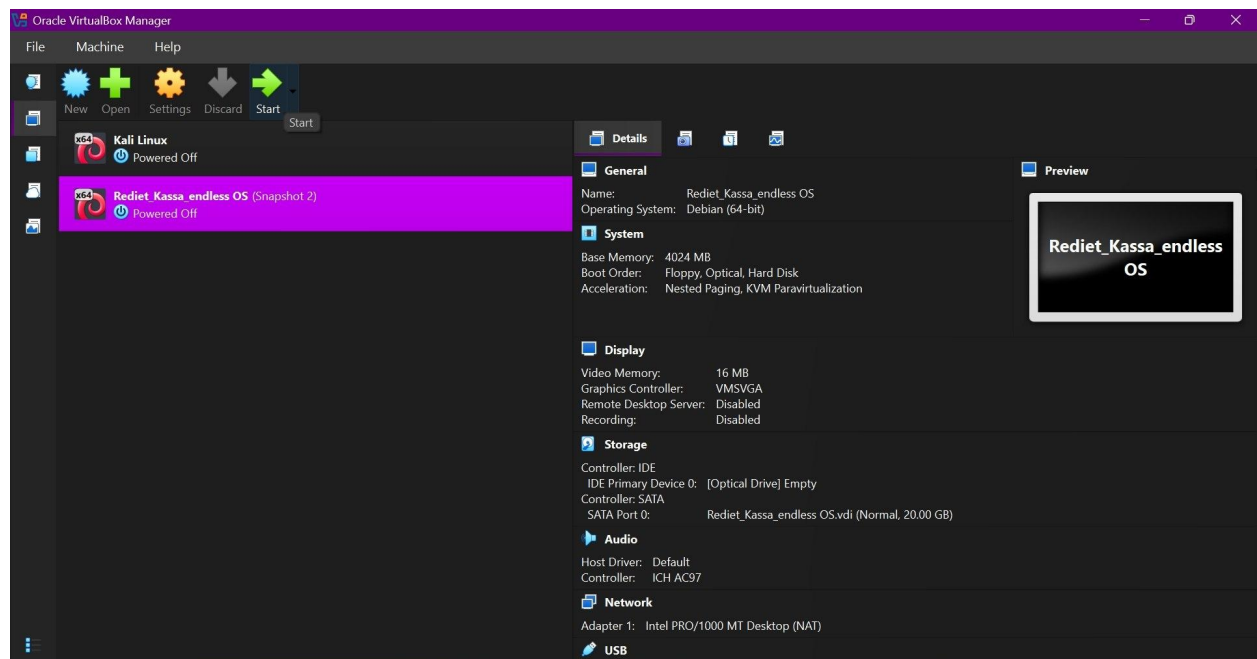
- File location and size: 40 GB – 60 GB (VDI, Dynamically allocated recommended).



6. Click Finish to create the virtual machine.

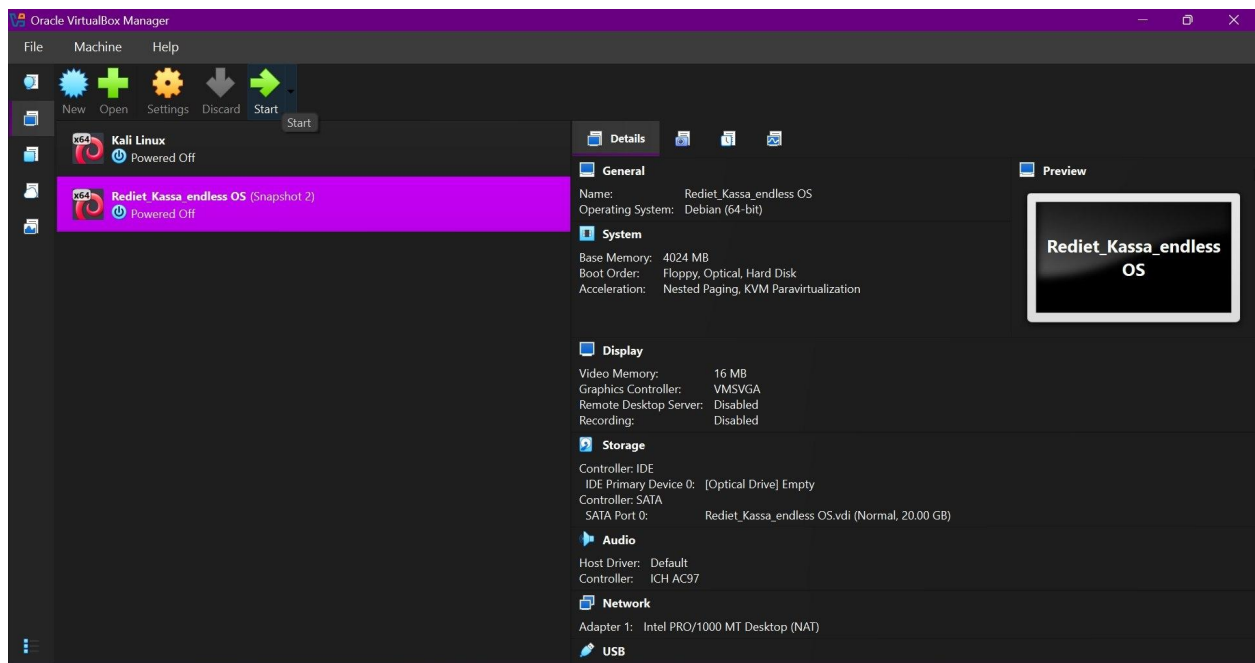
iii. Configure VM Settings (Before Starting)

7. Select the newly created VM → Click Settings.



8. Go to Storage, Under Controller: IDE, click the empty optical drive → Click the disk icon → Choose a disk file → Select the Endless OS ISO file.

9. Go to System → Processor tab → Enable PAE/NX and allocate at least 2 CPU cores



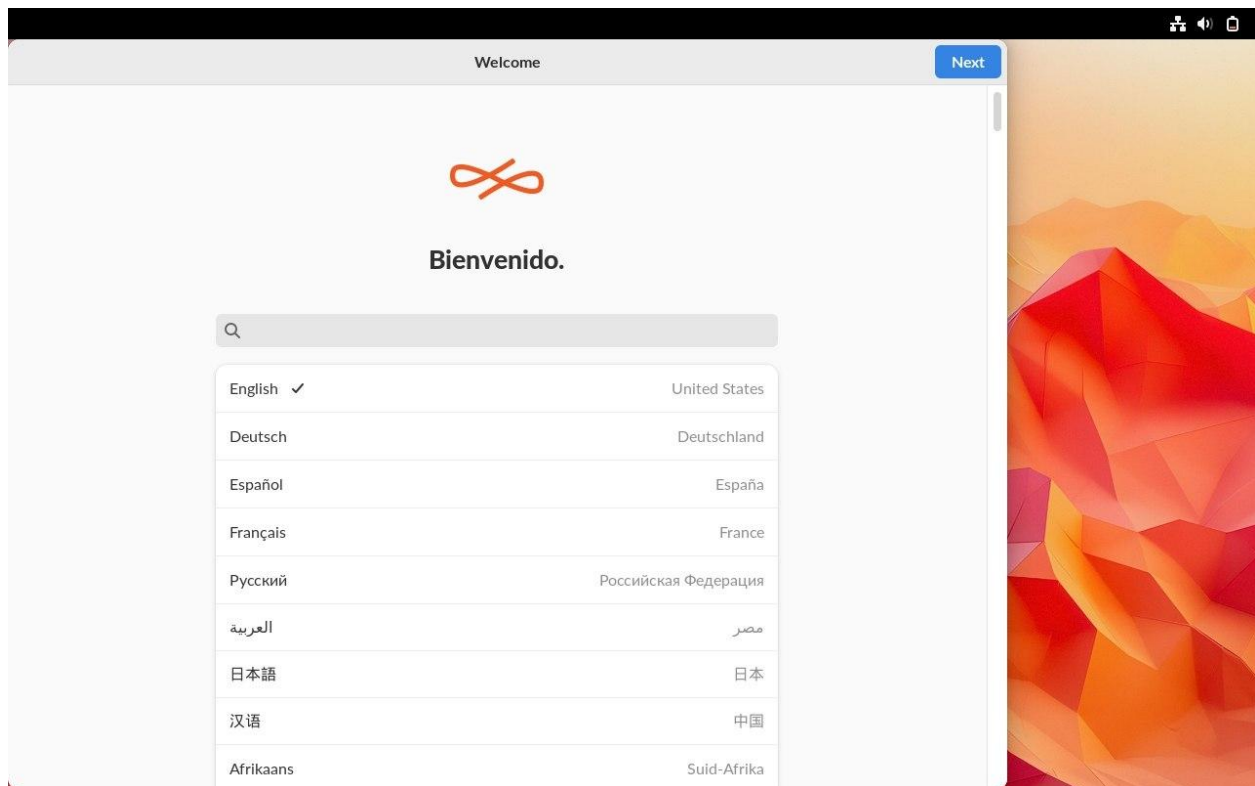
10. Go to Display→ Set Video Memory to 128 MB and enable 3D Acceleration

iv. Start Installation

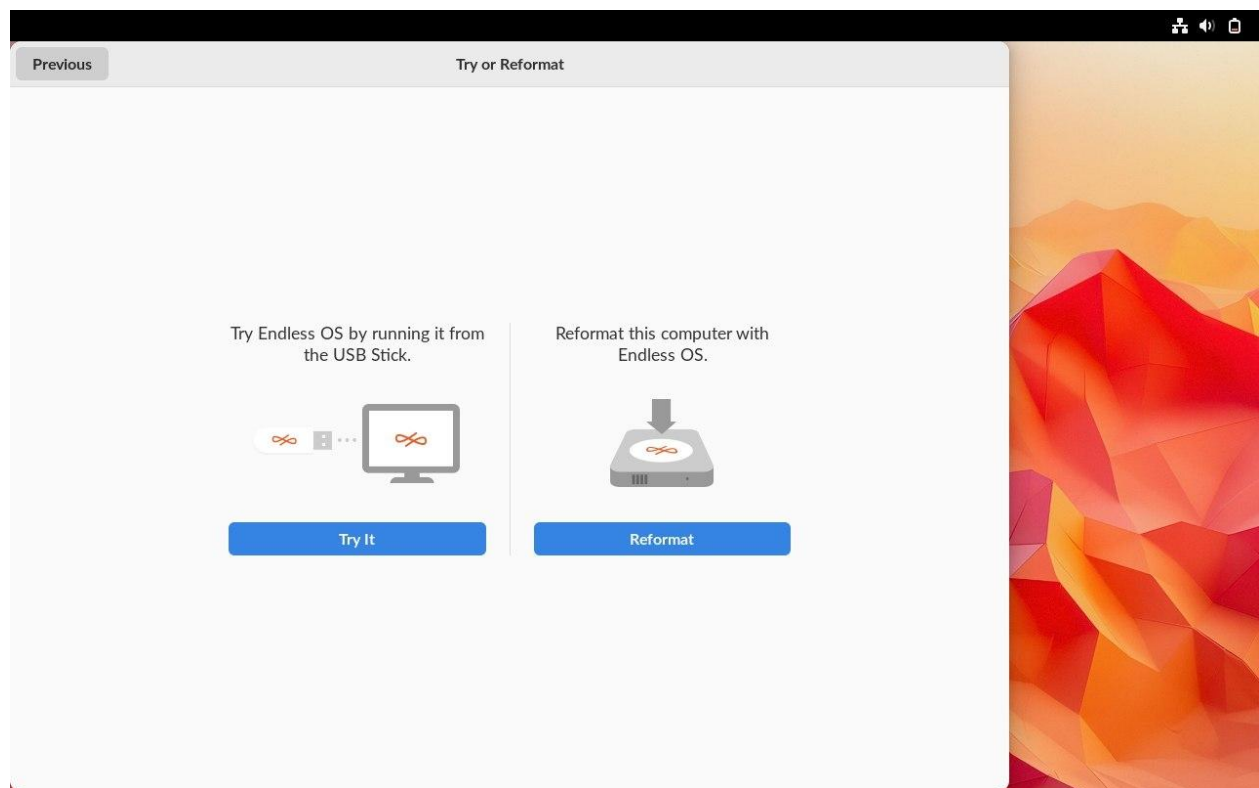


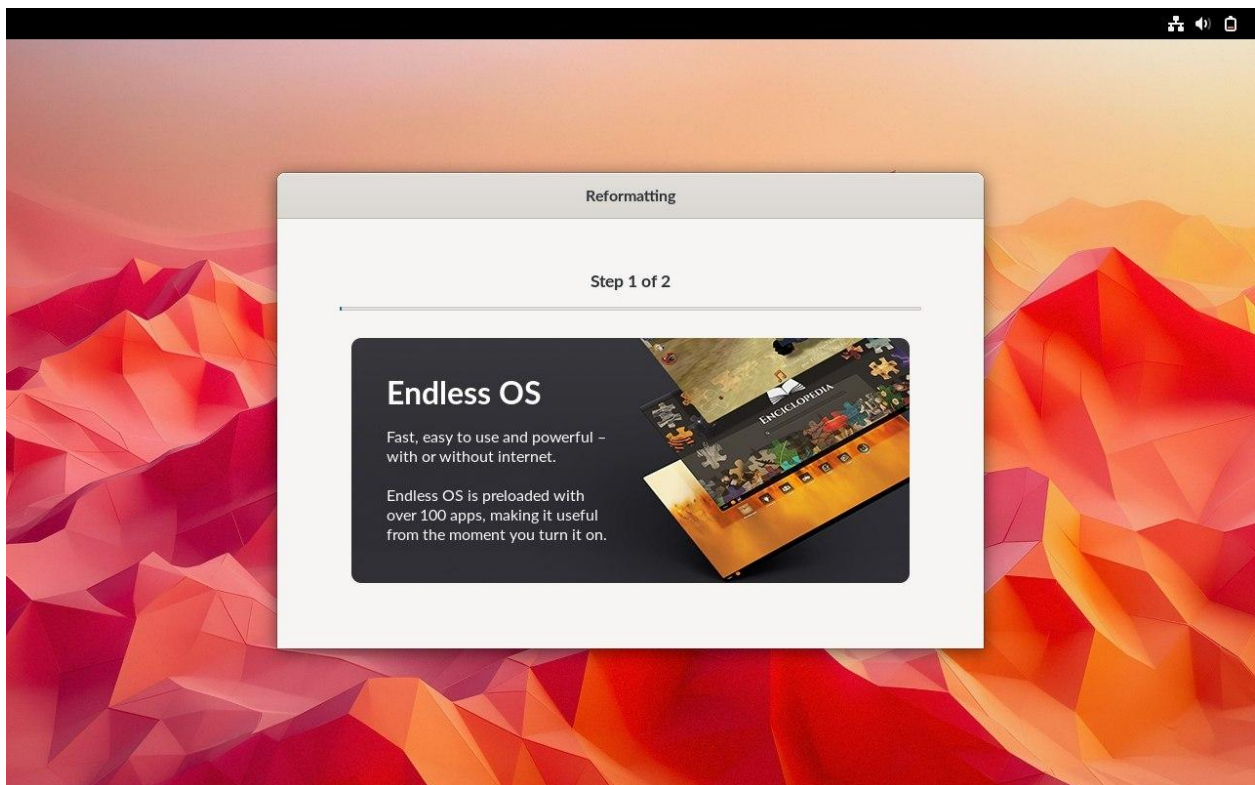
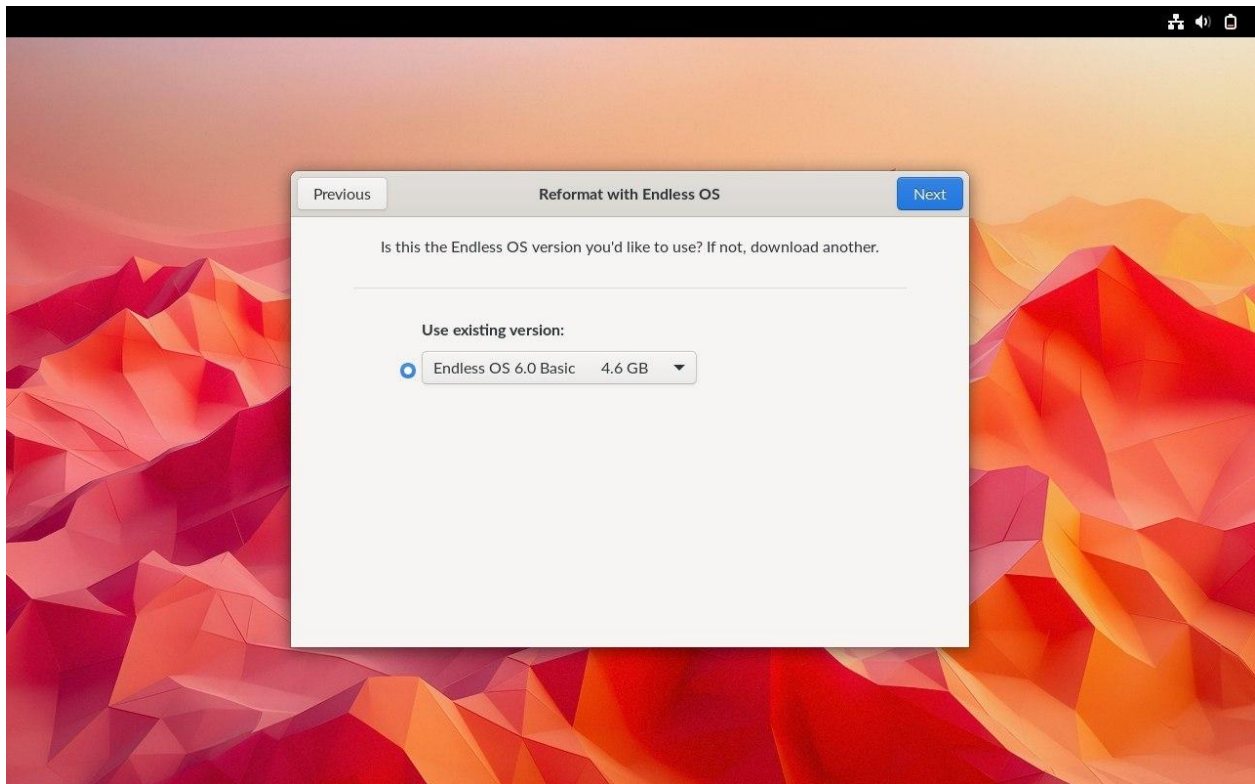
11. Select the VM and click Start.

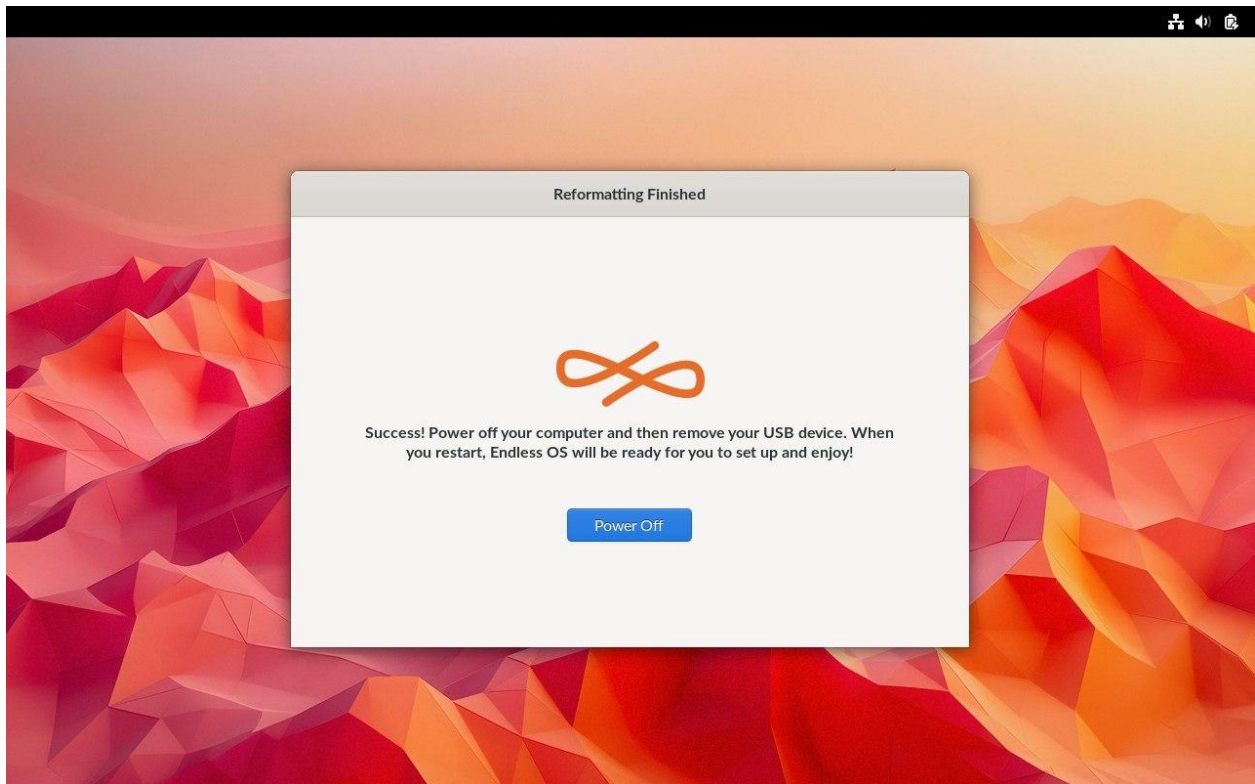
12. The Endless OS installer will launch. Follow the on-screen instructions: -Choose language



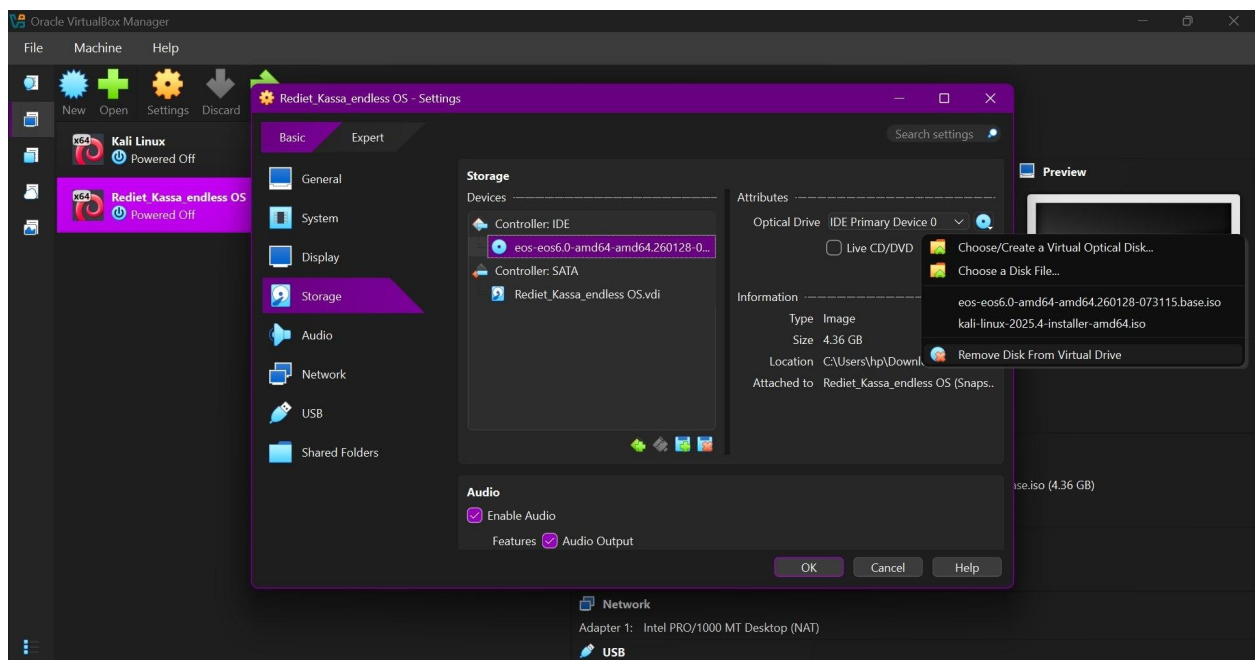
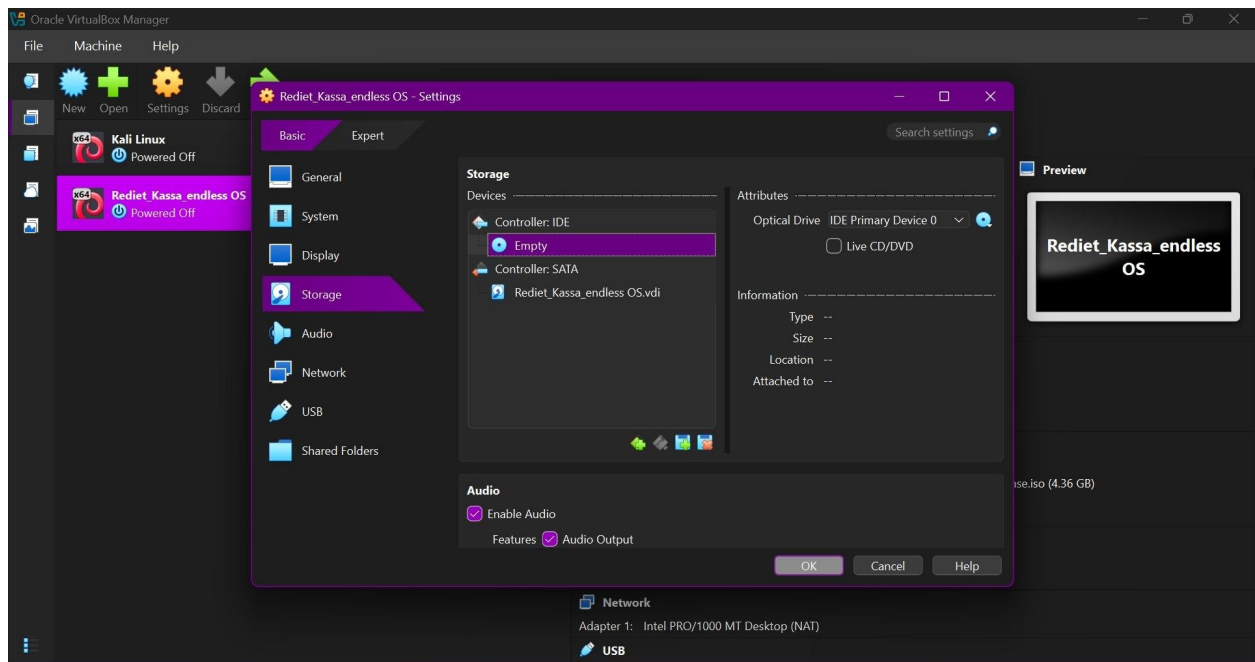
- Select keyboard layout
- Choose Erase disk and install Endless OS (safe inside VM)

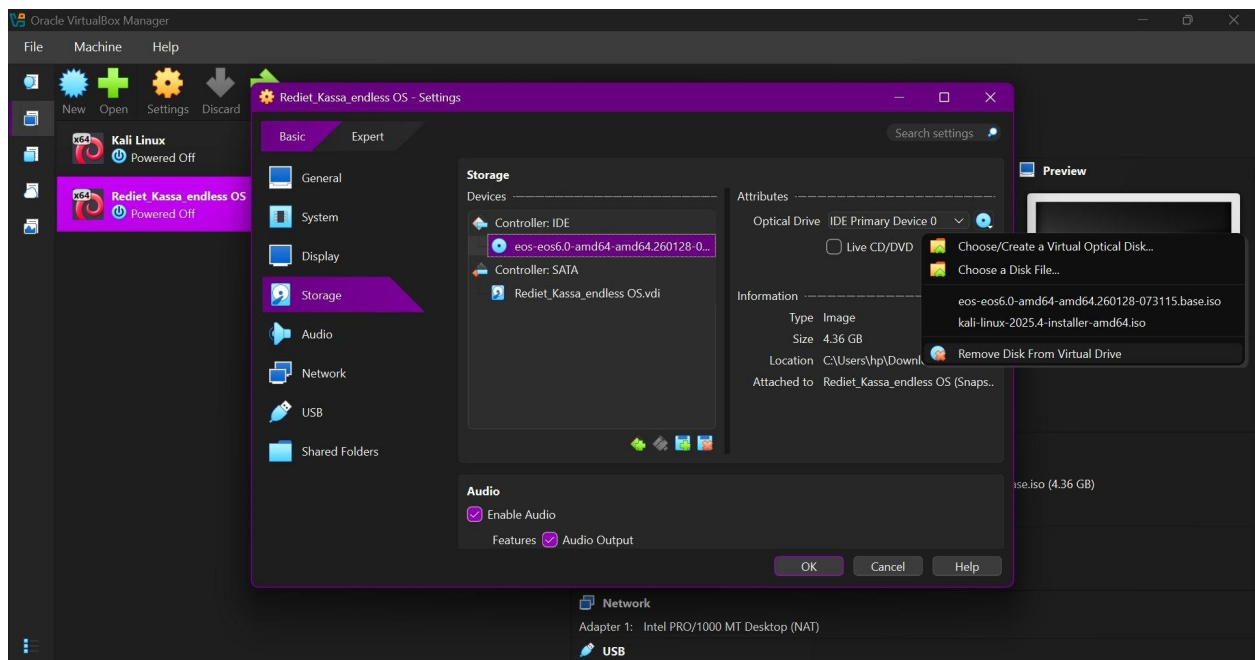
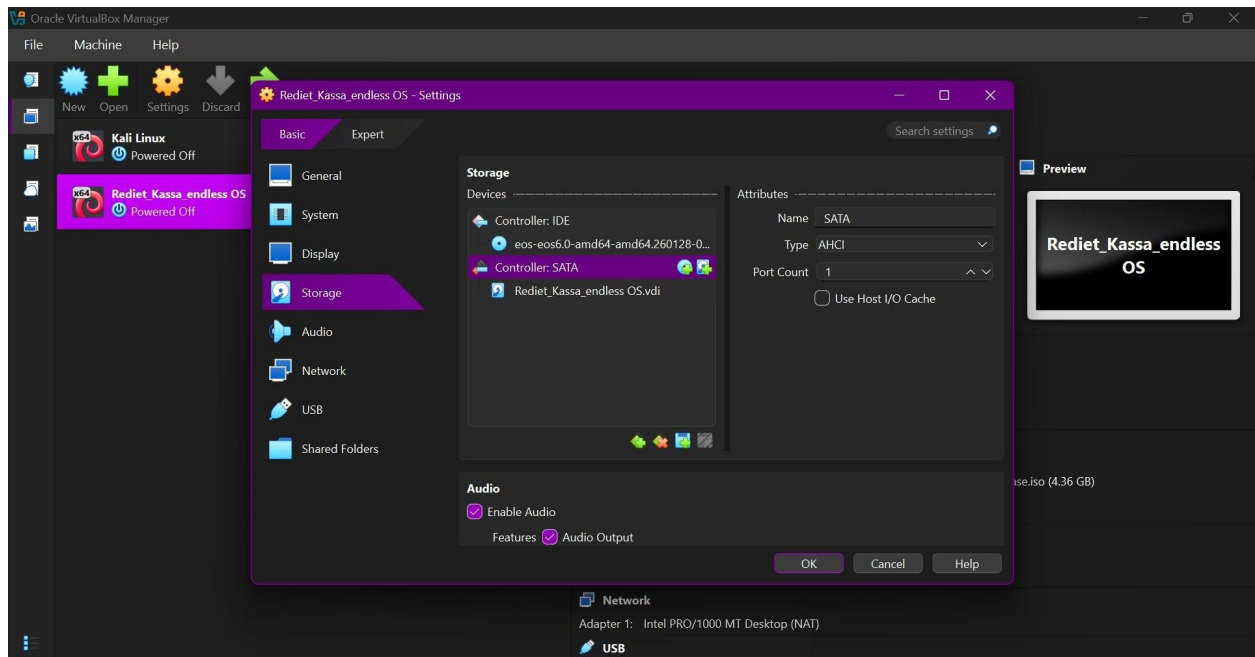


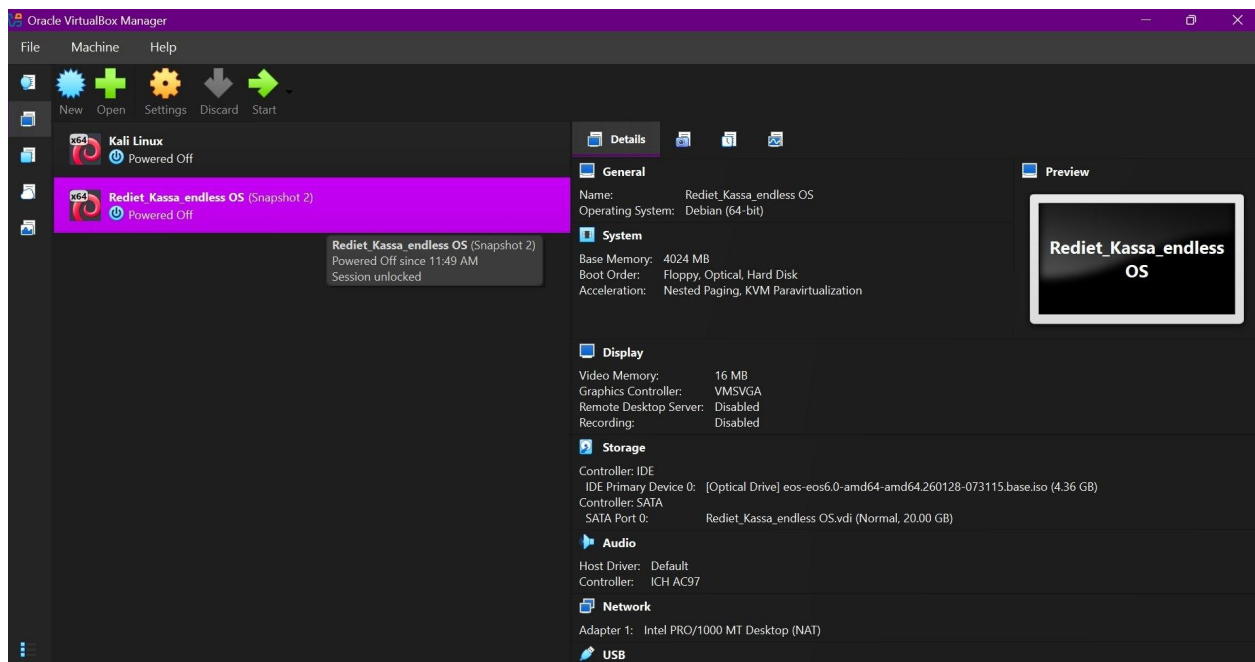
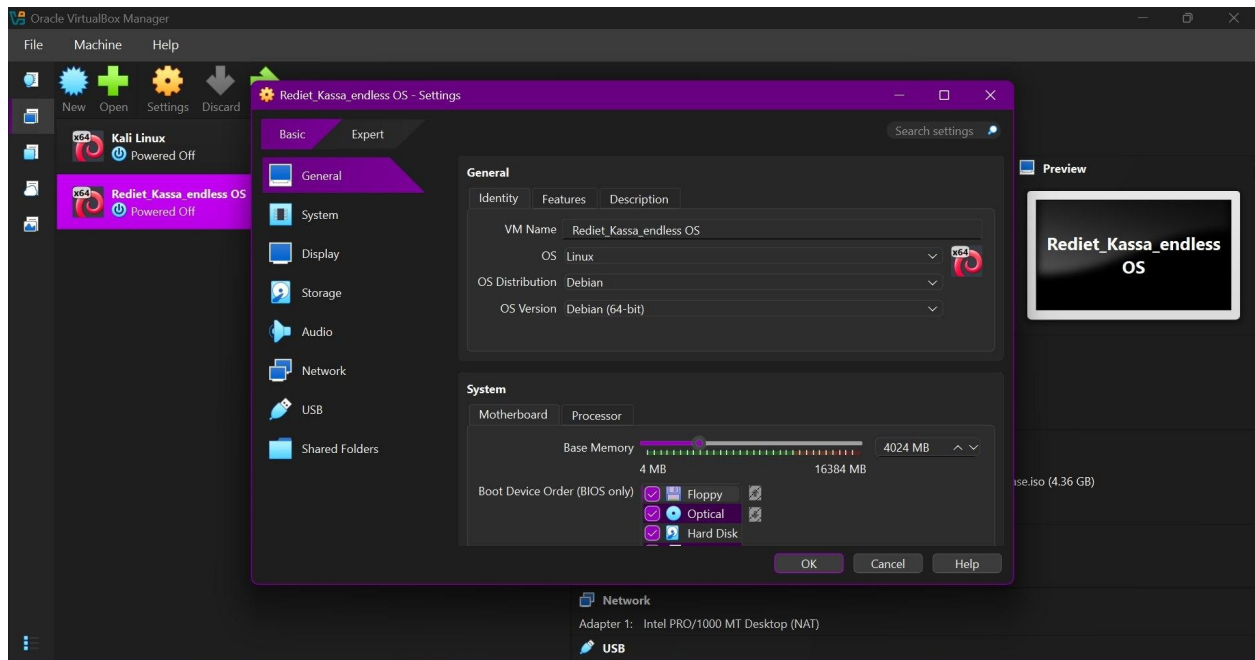




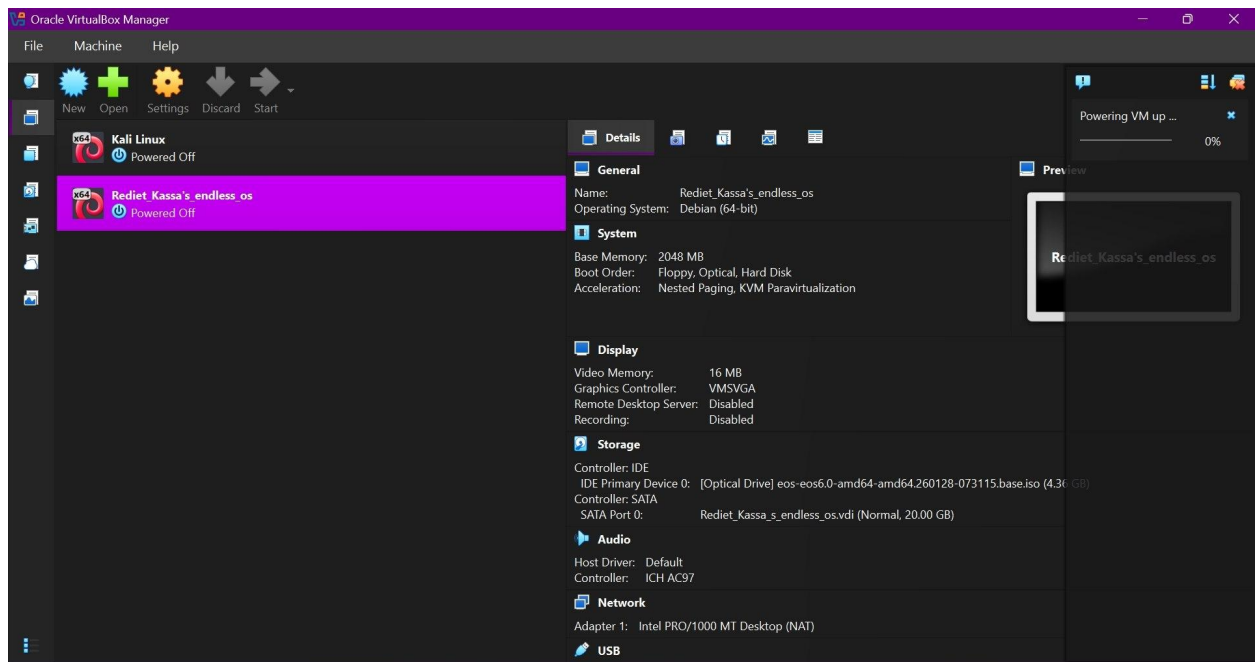
13. Create user account using your full name.







14. Wait for the installation to complete.



15. When prompted, restart the virtual machine. Remove the ISO from the virtual drive if asked.

File system Support

Endless OS is based on Debian GNU/Linux and uses ext4 as the default filesystem.

Supported filesystems

- NTFS– Read/Write
- FAT32– Read/Write
- exFAT– Read/Write
- ext4– Default (best performance and reliability)
- Btrfs– Supported
- ZFS– Supported with additional packages
- HFS+– Read/Write

- APFS – Limited read support

Why ext4

It provides excellent performance, journaling for data integrity, support for large volumes, and is the most stable and widely used Linux filesystem.

Advantages and Disadvantages

Advantages

- Very user-friendly and attractive interface
- Hundreds of educational and productivity apps pre-installed (works offline)
- Stable and secure (Linux-based)
- Low hardware requirements
- Ideal for beginners and education

Disadvantages:

- Fewer customization options than traditional Linux distributions
- Software availability relies heavily on Flatpak
- Smaller community compared to Ubuntu or Fedora
- Some advanced administrative tools are not easily accessible

Conclusion

The installation of Endless OS using Oracle VM VirtualBox was completed successfully. The process demonstrated how easy it is to install and test a modern Linux distribution in a safe virtual environment. Endless OS provides an excellent experience for users seeking simplicity and beauty in an operating system.

Future Outlook / Recommendation

Endless OS is highly recommended for students, educational institutions, and users with low-spec computers. In the future, I recommend installing Endless OS on real hardware (bare metal) to experience its full potential. It can also serve as a great secondary or daily driver OS alongside Windows.

Briefly explain the what, why, and how of virtualization in modern operating systems.

Virtualization

What is Virtualization

Virtualization is the process of creating a software-based (virtual) version of computing resources such as an operating system, server, storage, or network. In this project, we created a virtual machine that runs Endless OS on top of our host operating system using Oracle VM VirtualBox.

Why Virtualization

- Provides strong isolation between guest and host OS for safety.
- Allows running multiple operating systems on a single physical computer.
- Enables safe testing and experimentation without risking the host system.
- Reduces hardware costs and power consumption.
- Supports easy snapshots, cloning, and backups of entire OS environments.
- Extremely useful for education, development, and system administration training.

How does Virtualization Work

A Hypervisor (Virtual Machine Monitor) is the core software that enables virtualization.

- Type-1 Hypervisor (Bare-metal): Runs directly on hardware (e.g., VMware ESXi).
- Type-2 Hypervisor (Hosted): Runs on top of a host operating system (e.g., Oracle VM VirtualBox, VMware Workstation).

Key Mechanisms:

- Hardware-assisted virtualization (Intel VT-x / AMD-V)
- CPU, memory, and I/O virtualization
- Virtual hard disks, network adapters, and graphics cards
- Guest Additions / Tools for better integration (shared clipboard, better resolution, mouse integration, etc.)

In this project, I used Oracle VM VirtualBox, a popular free Type-2 hypervisor, which made the installation of Endless OS straightforward and efficient.